

Design Optimization and Harmonic Analysis of an Impeller

Impellers are critical components in turbomachinery systems used to pump fluids, circulate coolants, or compress gases in applications ranging from automotive water pumps and aerospace fuel systems to chemical process and hydraulic machinery. During operation, impellers experience combined thermal, pressure, and inertial loads and are subjected to periodic fluid-structure interactions as each blade passes through the surrounding flow field. This periodic excitation, known as the blade-passing frequency (**BPF**), can cause vibration or resonance if it aligns with the natural frequencies of the impeller. In this project, you will act as the design engineer tasked with optimizing an impeller's geometry and evaluating its dynamic performance under operating conditions. Your goal is to identify a robust design that maintains acceptable stress levels, avoids resonance, and minimizes harmonic deformation, all using ANSYS Workbench DesignXplorer.

The main objectives are to (1) perform coupled thermal-structural-modal-harmonic analyses of an impeller under realistic operating conditions, (2) explore the influence of geometric and operating parameters (blade height, impeller base thickness, speed, pressure, temperature, and blade count) on stress, stiffness, and vibration behavior, (3) quantify harmonic response due to blade-passing excitation and evaluate resonance avoidance through separation margins and, (4) apply design of experiments (DOE) using ANSYS DesignXplorer to identify trends and sensitivities, mirroring how design teams iterate industrial impeller concepts.

Baseline Design and Variables

- Diameter: 200mm
- Blade thickness $T_{\text{blade}}=3\text{mm}$
- Blade count: $Z=5$ and $Z=10$ (two variants)
- Material: Structural steel
- Fluid: Water $\rho=1000\text{kg/m}^3$

Design variables (for DOE):

- Blade height $H_b = 10\text{ mm}-30\text{ mm}$
- Impeller base thickness $T_{\text{base}}=3\text{ mm} - 6\text{ mm}$
- Rotational speed $N=4000-8000\text{ RPM}$
- Mean static pressure $P_{\text{mean}} = 1.5-2\text{MPa}$
- Blade temperature $T=80-140\text{ }^\circ\text{C}$

A Simplified impeller geometry showing key parameters for a 10-blade impeller is shown in **Figure 1**.

Harmonic Excitation: Each blade passage produces a periodic pressure fluctuation. The following relations can be used to determine the harmonic load P' and blade-passing frequency

$$f_{\text{BPF}} = Z \frac{N}{60}, \quad P' = C_p' \frac{1}{2} \rho U_{\text{tip}}^2, \quad U_{\text{tip}} = \frac{\pi D N}{60}, \quad \text{Assume } C_p' = 0.05$$

Resonance Avoidance: Show that your first elastic mode satisfies this requirement.

$$\text{Separation margin} = \frac{|f_n - f_{\text{BPF}}|}{f_{\text{BPF}}} \geq 0.2$$

It is recommended to carry out two smaller DOEs ($Z=5$, $Z=10$). Make assumptions on the realistic BCs for thermal analysis, including identifying the convection film coefficient.

Project Report Format

1: Introduction

- Any relevant background information emanating from articles featuring parameterization, impeller design, and blade-passing frequency (BPF). Cite at least two relevant articles.
- Problem statement

2: Approach/Methodology

- State assumptions used.
- Include all major simulation procedures, (i.e., BCs, etc.)

3: Results and Discussion

- Thermal, static, modal, and harmonic contour plots
- DOE matrix showing input and output parameters, charts from DOE, selected mode shapes, von Mises stress, and deformation contour map for the design chosen,
- A brief account of which variables control stiffness, frequency, and harmonic response, and how does blade count affects resonance risk.
- Etc.

4: Conclusions References

Files to be submitted: PDF file (project description) and CAD models (STEP file)

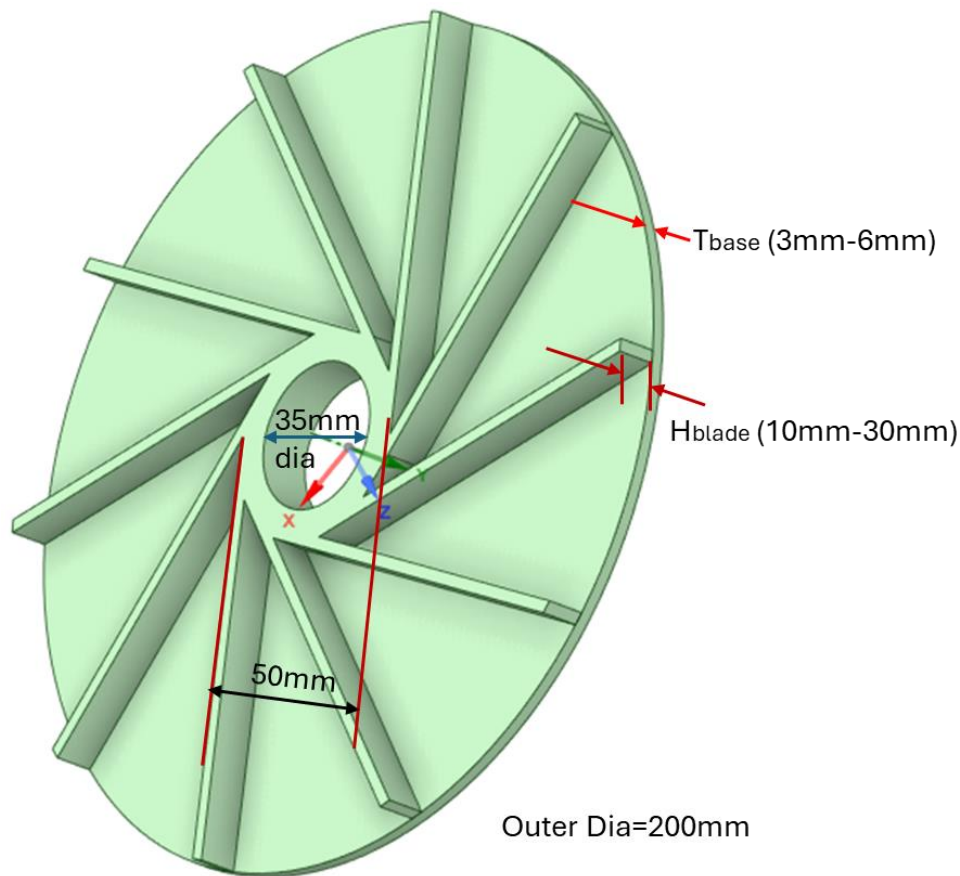


Figure 1. Simplified impeller geometry showing key parameters for a 10-blade impeller.

Below is a short YouTube video link on Parameterization using SpaceClaim

<https://www.youtube.com/watch?v=1gRYQEgFWDU>